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Voxelwise two-stage-clustering and machine-learning model for robust lung cancer tumor high-risk subregion segmentation on FDG PET

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INTRODUCTION

- A two-stage tumor subregion identification and segmentation model based on FDG PET of 23 patients with locally advanced non-small cell lung cancer
- Robust identification, segmentation, and monitoring of tumor subregions contribute to personalized radiation dose escalation and de-escalation strategies
- Automatic predictions during radiation therapy on FDG PET/CT predicts outcomes in locally advanced NSCLC patients

AIMS

- **To identify and segment risky regions in lung cancer tumor with neural network**
- **To provide precision radiotherapy treatment strategies with automated machine learning methods for 23 patients with locally advanced non-small cell lung cancer**
- **To monitor lung cancer tumor target responses and class responses in low risky subregions**

KEY FINDINGS

- Voxel clusters without spatial constraint achieved higher statistical independence than those with equally weighted SUV and voxel position (SC 0.61 [0.55-0.68] vs. 0.30 [0.23-0.47], CH-index 9122 [865-102062] vs. 3966 [312-49227]).
- Voxel clusters were more stable between PETpre and PETmid using the two-stage clustering method, with DS of 0.93 [0.58-1.00] and ED of 0.17 [0.01-2.78] in the MTV 2 Hi-cluster compared to the benchmark with optimal weights (DS 0.75 [0.50-0.92] and ED 1.16 [0.31-3.18]).

METHODS

FDG-PET/CT imaging

- Data: the pre-treatment (PETpre) and three-week mid-treatment fluorodeoxyglucose (FDG) PET imaging (PETmid) of 23 patients with locally advanced non-small cell lung cancer who received chemoradiotherapy
- Pre-RT: median 1 week prior, range < 1 month prior to 1st radiotherapy fraction
- Week 3 mid-RT: median 24 Gy in 12 fractions, range 20-26 Gy in 10-13 fractions
- Standardized acquisition and reconstruction imaging protocol using cross-calibrated PET/CT scanners
- Primary metabolic tumor volumes (MTV) semi-automatically defined by PET EdgeTM (MIM Software, Cleveland OH)
- Rigid alignment of primary MTV between planning CT, pre-RT PET/CT, mid-RT PET/CT, and RT dose
- Co-registered PET voxel grids resampled to isotropic 2 mm dimensions

Two-stage-clustering diagnosis and threshold variation prognosis

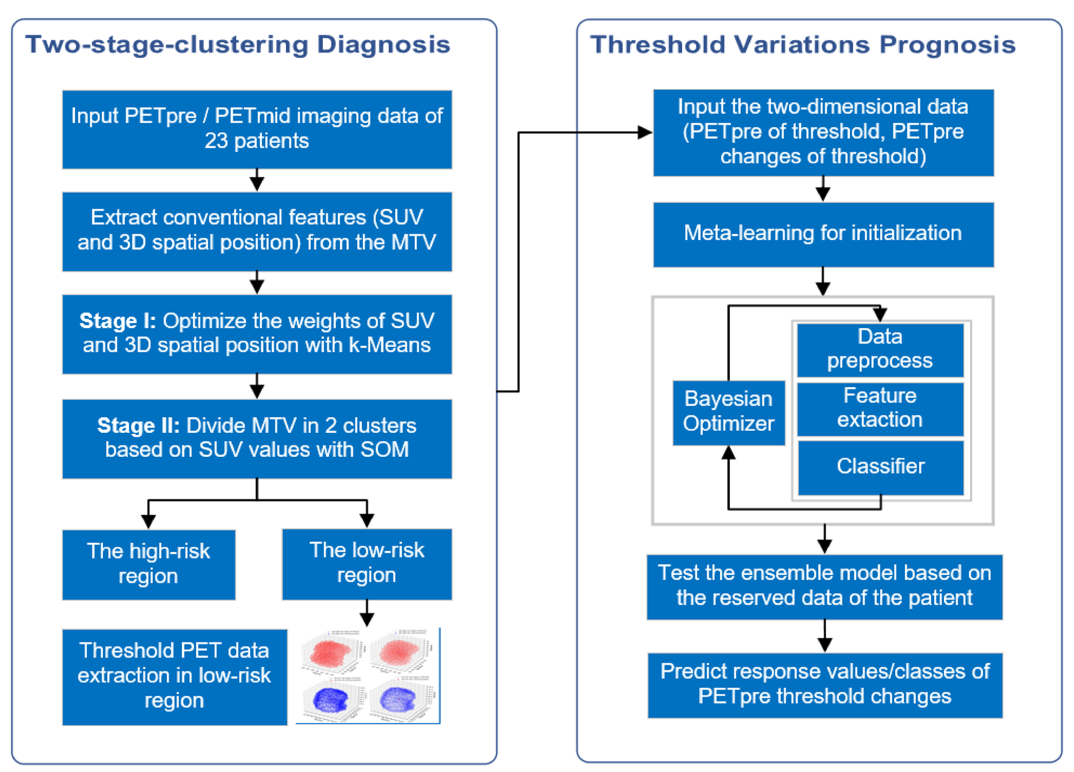

Figure 1. The working-flow frame of the model

STEP I: Weight optimization. In the first stage, we measured the statistical independence of PETpre and PETmid voxel clusters with and without spatial constraints. We compared equal and optimal weights of voxel features (standardized uptake value SUV, 3D position) by k-Means [1], where cluster separability was evaluated by Silhouette Coefficient (SC) and Calinski-Harabasz-index (CH-index).

STEP II: Clustering based on SOM. In the second stage, to reinforce clustering stability, k-Means clusters were ingested into a Self-Organizing Map (SOM) [4] and similarity between PETpre and PETmid clusters was optimized. On-treatment cluster stability using Dice similarity (DS) of cluster contours was iteratively maximized and Euclidean distance (ED) of cluster centroids was iteratively minimized by updating SOM network nodes.

Two-stage-clustering diagnosis: tumor clustering

- MTV clustering algorithms: k-Means (Stage I), Self Organizing Map (SOM, Stage II)
- Voxel similarity defined by Euclidian distance metric: root-mean-squared z-scores of FDG PET SUV and 3D spatial position
- Cluster definitions: 2 High / Low clusters, 3 High / Medium / Low Clusters
- Cluster performance metric: Silhouette Coefficient (SC), Calinski-Harabasz-index (CH-index)

Threshold variation prognosis: prediction modeling

- Hierarchical sensitivity analysis for predicting regional tumor response on week 3 mid-RT FDG PET from baseline
- Pre-RT semantic features: SUVmax [2], ΔSUVmax which can be calculated by (PETmid SUVmax - PETpre SUVmax)/PETpre SUVmax
- Logistic & non-linear classifiers to predict ΔSUVmax class (binary definition of PET responder class: ΔSUVmax ≥ 30% decrease)
- Regressor and classifier algorithms ranking: Automated machine learning algorithms supported by Auto-sklearn [3,5]

RESULTS

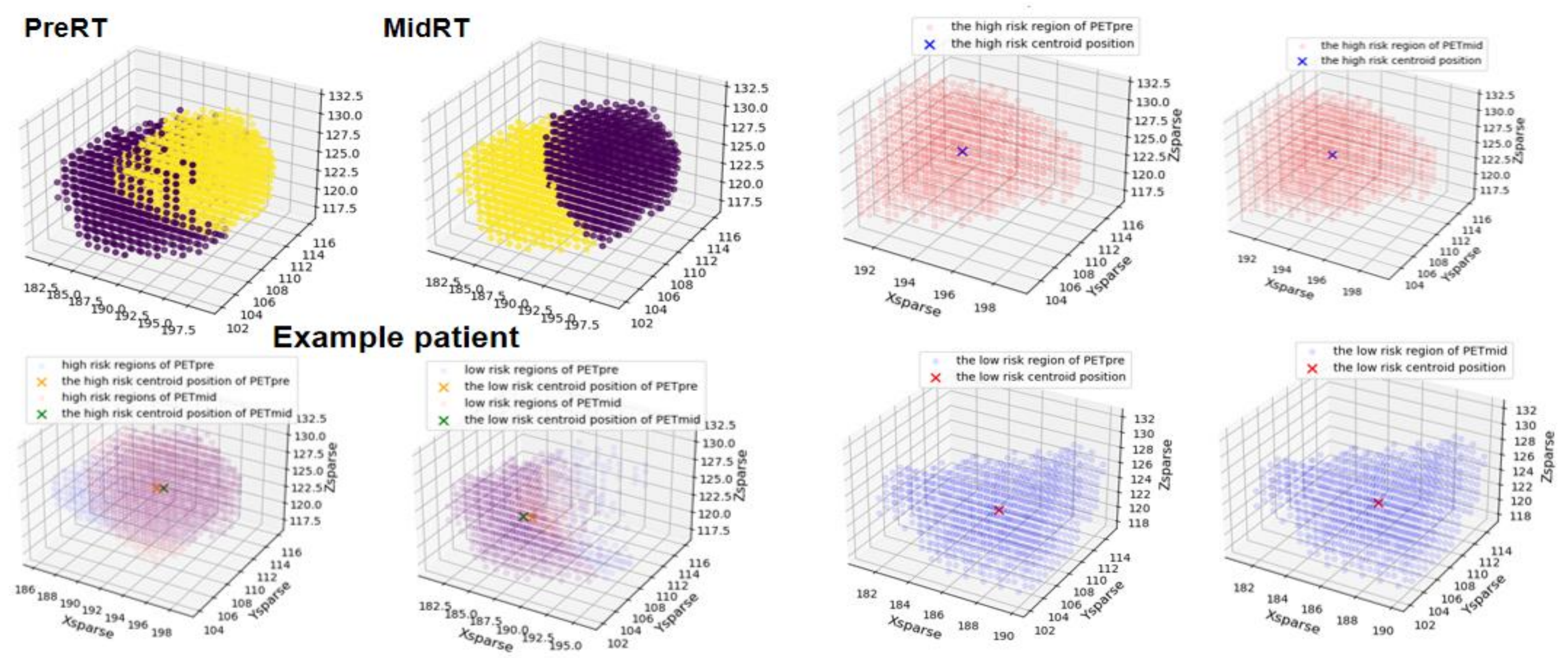

Figure 1. Tumor spatial clustering visualization plots and subplots of baseline (PreRT), 3-week response (MidRT) FDG PET/CT based on k-Means with equal weights and overlap imaging between MTVpre and MTVmid of high/low-risk regions during chemo-radiotherapy in one example patient with locally advanced non-small cell lung cancer. In Figure 1, the tumor of the example patient is segmented by k-Means with equal weights, which enforces spatial contiguity of voxel cluster positions by dividing the tumor into hemispheres.

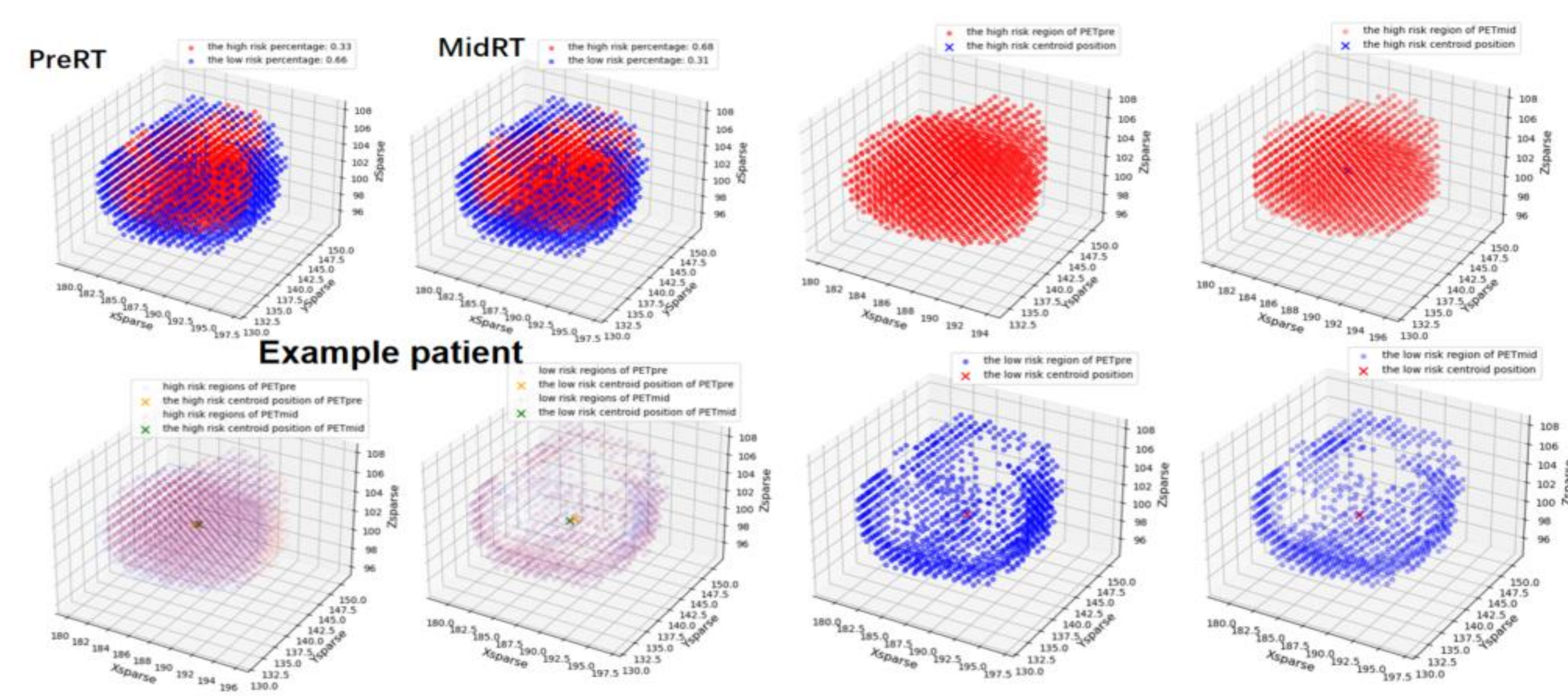

Figure 2. Tumor spatial clustering visualization plots and subplots of baseline (PreRT), 3-week response (MidRT) FDG PET/CT based on our two-stage clustering methods and overlap imaging between MTVpre and MTVmid of high/low-risk regions during chemo-radiotherapy in one example patient with locally advanced non-small cell lung cancer. MTV are clustered by the two-stage clustering method based on the SOM model, and results are shown in Figure 2, where visualized plots show that high-risk clusters are more centrally located in metabolic regions.

Table 1. Silhouette coefficient and CH-index of MTV based on k-Means with two-stage clustering method and k-Means with original weight [1], where evaluation values are consisted of median, minimum and maximum values.

Silhouette Coefficient	PETpre		PETmid	
	2 clusters	3 clusters	2 clusters	3clusters
k-Means (optimal weights)	**0.61 (0.55-0.68)**	**0.56 (0.53-0.59)**	**0.61 (0.55-0.72)**	**0.56 (0.53-0.67)**
k-Means (equal weights)	0.30 (0.23-0.47)	0.27 (0.21-0.35)	0.27 (0.22-0.36)	0.28 (0.22-0.30)
CH-index	**2 clusters**	**3 clusters**	**2 clusters**	**3clusters**
k-Means (optimal weights)	**9112 (865-102062)**	**11274 (1201-131032)**	**8671 (887-192015)**	**10821 (1153-240551)**
k-Means (equal weights)	3966 (312-49227)	3091 (304-40618)	3892 (292-71669)	3167 (284-51583)

Table 2. Tumor voxel cluster spatial overlap and centroid migration evaluation results by DS and ED. ED and DS are calculated based on centroid positions of cluster pairs, which aims to evaluate clustering stability. In low-risk regions, MTV 2 Lo-cluster of two-stage clustering had higher overlap and smaller centroid migration, whose median values of DS is 0.09 higher than that of k-Means, and ED is 0.49 lower than that of k-Means. Therefore, we applied SOM as the second step of our two-stage clustering.

	MTV 2 Lo-cluster		MTV 2 Hi-cluster	
	k-Means	SOM	k-Means	SOM
Euclidean Distance	0.67 (0.15-0.46)	**0.18 (0.01-1.20)**	1.16 (0.31-3.18)	**0.17 (0.01-2.78)**
Dice Similarity	0.91 (0.81-1.00)	**1.00 (0.63-1.00)**	0.75 (0.50-0.92)	**0.93 (0.58-1.00)**

Table 3. LOOCV RMSE average evaluation results of threshold changes prediction in low-risk region by different regression model, where the input data is PETpre SUVmax and the prediction target is (PETmid SUVmax - PETpre SUVmax)/PETpre SUVmax.

Models	MAE	MSE	RMSE	MAPE	MedianAE
Auto-sklearn	**0.13**	**0.03**	**0.16**	**1.75**	**0.10**
SVR	0.15	0.03	0.18	1.67	0.14
DTR	0.19	0.05	0.21	1.56	0.18
RFR	0.17	0.04	0.19	1.53	0.17
ABR	0.17	0.04	0.19	1.47	0.17
GBR	0.19	0.04	0.21	1.55	0.18

REFERENCES

1. Duan C, Chaovalitwongse WA, Bai F, Hippe DS, Wang S, Thammasorn P, et al. Sensitivity analysis of FDG PET tumor voxel cluster radiomics and dosimetry for predicting mid-chemoradiation regional response of locally advanced lung cancer. Physics in Medicine and Biology. 2020;65:205007.
2. Horn KP, Thomas H, Vesselle HJ, Kinahan PE, Bowen SR. Reliability of Quantitative 18F-FDG PET/CT Imaging Biomarkers for Classifying Early Response to Chemoradiotherapy in Patients With Locally Advanced Non–Small Cell Lung Cancer. Clinical Nuclear Medicine. 2021;46:861.
3. Feurer M, Klein A, Eggensperger K, Springenberg JT, Blum M, Hutter F. Auto-sklearn: Efficient and Robust Automated Machine Learning. NIPS 2015: Proceedings of the 28th International Conference on Neural Information Processing Systems. CANADA; 2015. p. 2944-52.
4. Kohonen T. The Self-organizing Map. Proceedings of the IEEE. 1990;78:1464-80.
5. Kleesiek J, Kersjes B, Ueltzhoffer K, Murray JM, Rother C, Kothe U, et al. Discovering Digital Tumor Signatures-Using Latent Code Representations to Manipulate and Classify Liver Lesions. Cancers. 2021;13:3108.

CONCLUSIONS

- Two-stage voxel clustering defines statistically independent and spatially reliable subregions as compared to k-Means models with equal weights in the same example patient during treatment
- Robust multi-resolution clustering provides clinicians with support for response assessment and biologically adaptive radiation therapy targets
- Predict lung cancer tumor target responses and class responses of tumor subregions for patients with automated machine learning algorithms

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